

2025-2026

DISASTERGUARD

DISASTER GUARD

AI-Powered Disaster Risk Prediction
& Route Safety Platform

Empowering people with intelligent insights,
real-time alerts, and safer decisions
when it matters most.



DESIGNED
MAN

AI • Machine Learning
Real-Time Weather
Safe Route Analysis
Artificial Intelligence
Machine Learning
Full-Stack Development
Data Analytics

DisasterGuard is an AI-powered web application designed to improve disaster awareness through intelligent risk prediction, real-time weather monitoring, interactive mapping, and safer route planning. This case study explores the project's research, design process, technology stack, implementation, and development journey, demonstrating how modern technologies can be combined to address real-world challenges.

By Mansi Potwar

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ABSTRACT

DisasterGuard is an AI-powered web application designed to help users assess disaster risks, monitor real-time weather conditions, and identify safer travel routes. The platform combines machine learning, historical disaster datasets, interactive maps, and external APIs to provide location-based risk predictions and environmental insights.

Built using React.js, TypeScript, Python, Flask, and MongoDB Atlas, the application demonstrates the integration of modern web development with machine learning to create a responsive and scalable disaster management solution.

While the prediction model relies on historical data and serves as a decision-support tool rather than an absolute forecasting system, DisasterGuard showcases how AI and real-time data can improve disaster awareness, preparedness, and informed decision-making.



PROBLEM STATEMENT

Natural disasters can occur without warning, making it difficult for people to quickly access reliable information about disaster risks, weather conditions, and safe travel routes. Existing solutions often provide only weather updates or alerts, requiring users to switch between multiple platforms.

DisasterGuard addresses this challenge by combining AI-powered disaster prediction, real-time weather data, interactive maps, and safe route recommendations into a single, easy-to-use web application.



OBJECTIVES

Design an intelligent platform that helps users predict disaster risks, monitor weather conditions, and identify safer routes using AI and real-time data.

Deliver a modern, user-friendly web application that showcases the integration of machine learning, mapping services, and external APIs.

KEY FEATURES



AI Disaster Prediction

Predicts potential disaster risks for a selected location using machine learning models and historical disaster data.



Real-Time Weather

Provides live weather updates including temperature, humidity, wind speed, pressure, and other key environmental conditions.



Safe Route Planning

Finds and visualizes optimized and safer routes between locations using map-based navigation and routing services.



Interactive Maps

Explore locations, view risk levels, and interact with maps for better geographical context and situational awareness.



Email Alerts

Sends instant email notifications and alerts about potential disasters and critical weather updates.



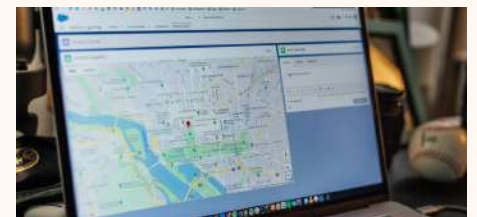
Responsive & User-Friendly

Built with modern web technologies to deliver a smooth, responsive, and intuitive experience across all devices.



AI-Powered Decision Support

Combines multiple data sources and AI insights to help users make informed and timely decisions.



TECHNOLOGY STACK

DisasterGuard is built using a modern and scalable technology stack that combines frontend, backend, machine learning, database, and third-party APIs to deliver accurate disaster predictions and real-time insights.

	FRONTEND DEVELOPMENT	React.js, TypeScript, Vite, Tailwind CSS
	BACKEND DEVELOPMENT	Python, Flask, REST API
	MACHINE LEARNING	Scikit-learn, Pandas, NumPy
	DATABASE	MongoDB Atlas
	APIs & SERVICES	OpenWeather API, Google Maps API
	DEVELOPMENT TOOLS	VS Code, Git, GitHub, npm, Node.js

SYSTEM ARCHITECTURE

DisasterGuard follows a modular architecture that integrates a modern web frontend, a Python-based backend, machine learning models, external APIs, and a cloud database. Each component works together to process user requests, generate disaster risk predictions, and deliver real-time weather, mapping, and route information through a unified interface.

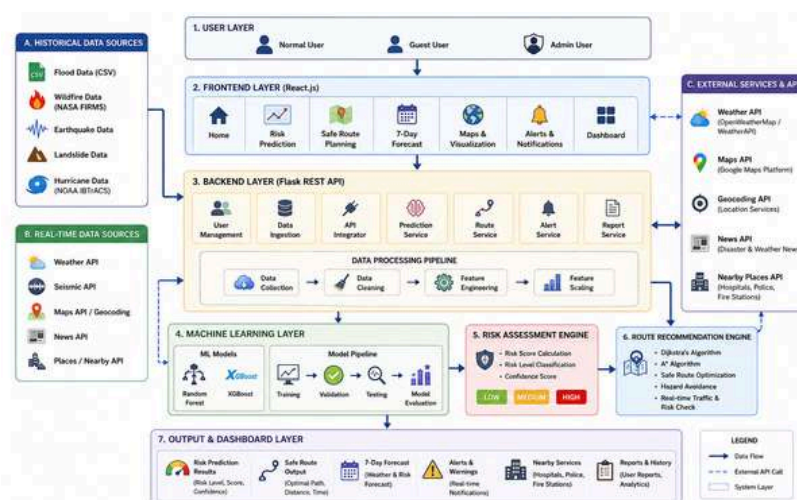
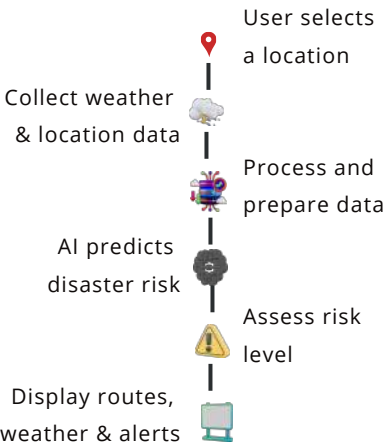


Figure 1 — Overall System Architecture of the DisasterGuard Platform

Architecture Components

- **User Layer** – Users interact with the application through a responsive web interface.
- **Frontend** – Built with React.js and TypeScript to provide an intuitive and interactive user experience.
- **Backend** – Processes requests, manages application logic, and communicates with the machine learning model and external APIs.
- **Machine Learning** – Uses Random Forest and XGBoost models to estimate disaster risk based on historical data.
- **External APIs** – Integrates weather, geolocation, mapping, and route services to provide real-time information.
- **Database** – MongoDB Atlas stores application data and supports efficient data management.
- **Output** – Displays disaster predictions, weather updates, maps, route recommendations, and email alerts.

PROPOSED APPROACH



DisasterGuard follows a structured workflow that combines user input, machine learning, real-time APIs, and interactive visualization to deliver disaster risk predictions and safety recommendations. The system processes both historical disaster data and live environmental information to provide meaningful insights through a single web application.

Workflow

1. User Input: The user enters a location to initiate disaster risk analysis.
2. Data Collection: Historical disaster datasets and real-time data are collected from weather, mapping, and geolocation APIs.
3. Data Processing: The collected data is cleaned, prepared, and transformed for machine learning analysis.
4. AI Prediction: Random Forest and XGBoost models analyze the processed data to estimate disaster risk.
5. Risk Assessment: The prediction is categorized into different risk levels to support informed decision-making.
6. Results & Visualization: The application displays disaster predictions, weather information, interactive maps, safe routes, and email alerts through a user-friendly dashboard.

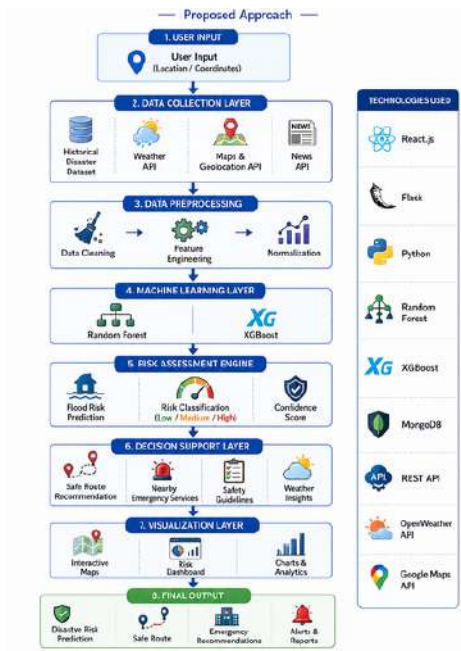


Figure 2 — Proposed Workflow of the DisasterGuard System

AI Engine & Data Integration

The intelligence behind DisasterGuard is powered by machine learning models trained on historical disaster datasets and enhanced with real-time environmental data. By combining AI predictions, external APIs, and cloud-based storage, the system delivers accurate, location-aware disaster insights through a unified platform.

© EARLY DISASTER INTELLIGENCE PLATFORM

Detect Disaster Risk Before It Strikes



Machine Learning

The prediction engine uses Random Forest and XGBoost models trained on historical datasets covering floods, earthquakes, landslides, wildfires, and hurricanes. Before prediction, the data is cleaned, processed, and transformed through feature engineering and normalization to improve model performance. Based on the analysis, the system classifies disaster risk into Low, Medium, or High categories.



External APIs

To provide real-time insights, DisasterGuard integrates OpenWeather for current weather conditions, OpenMeteo for weather forecasts, Google Maps for interactive map visualization, OpenRouteService (ORS) for route planning, and a Geocoding API to convert user-entered locations into geographic coordinates.



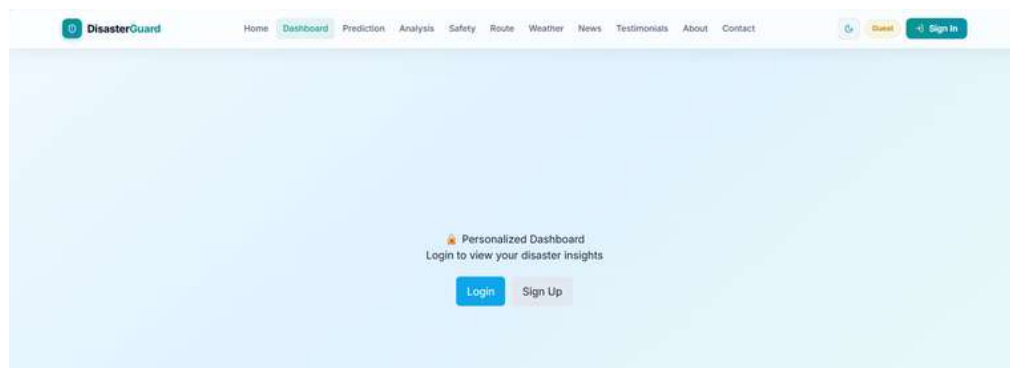
Database

MongoDB Atlas is used as the cloud database to securely store application data, historical disaster datasets, user information, and prediction records, ensuring efficient and scalable data management.

EXPERIENCE THE PRODUCT

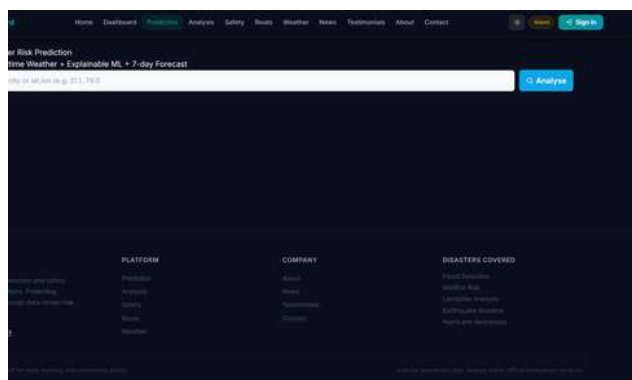
A visual walkthrough of the application's core interface and user experience.

Home Dashboard



The central dashboard provides quick access to disaster prediction, weather monitoring, interactive maps, and route planning through a clean and intuitive interface.

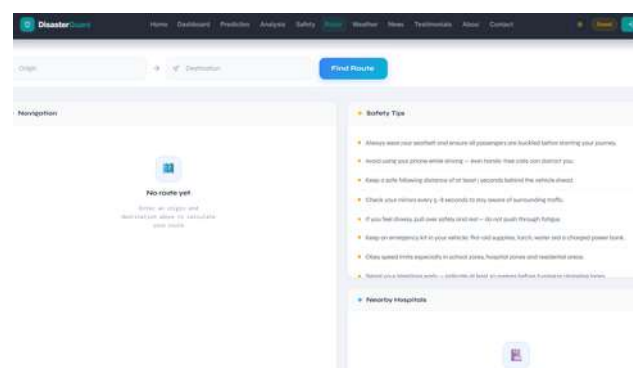
AI Risk Prediction



Users can search for a location and generate AI-powered disaster risk predictions based on historical datasets and environmental conditions.

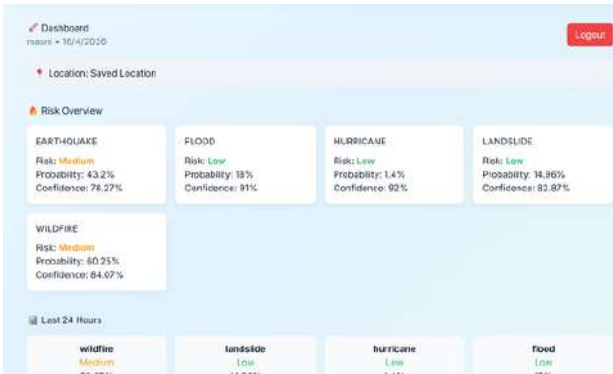
Safe Route Planning

Generates an optimized route between the selected locations while visualizing the path on an interactive map to support safer travel and better navigation.



COMPLETE APPLICATION WALKTHROUGH

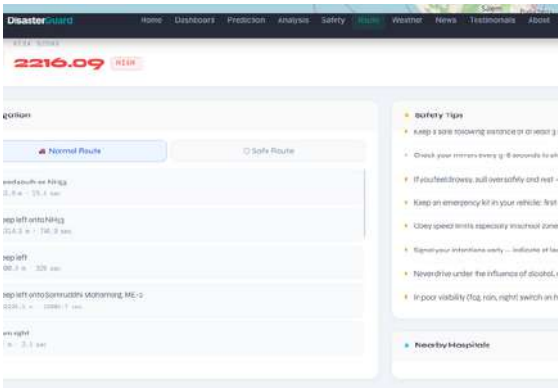
Explore the key modules of DisasterGuard.



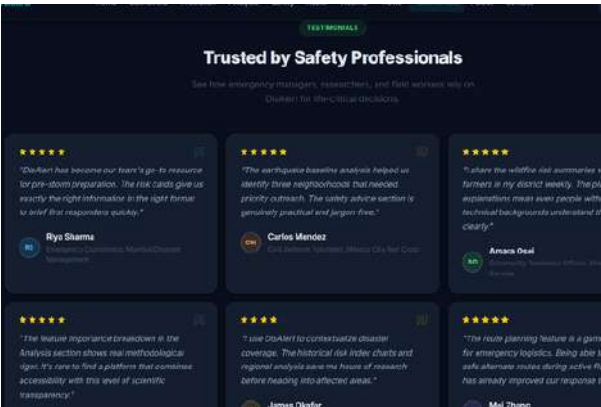
Dashboard

Find optimized routes with interactive map-based navigation.

A centralized dashboard providing quick access to all DisasterGuard features.



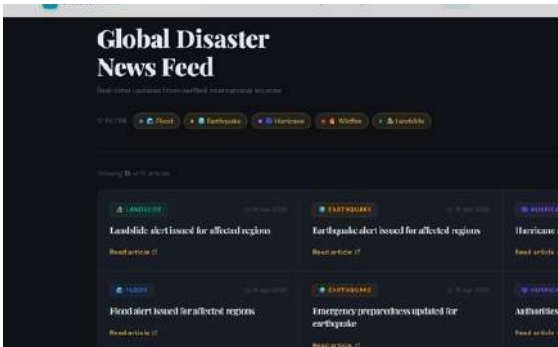
Route



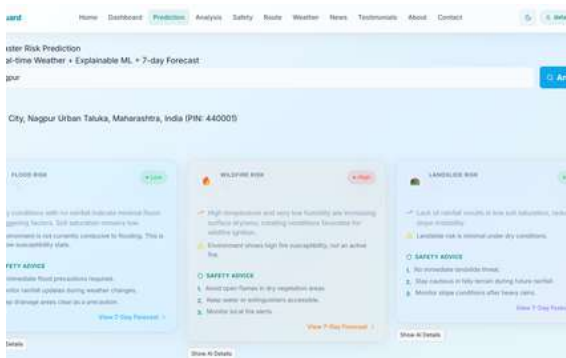
Testimonials

Stay informed with the latest disaster-related news and updates.

A centralized dashboard providing quick access to all DisasterGuard features.

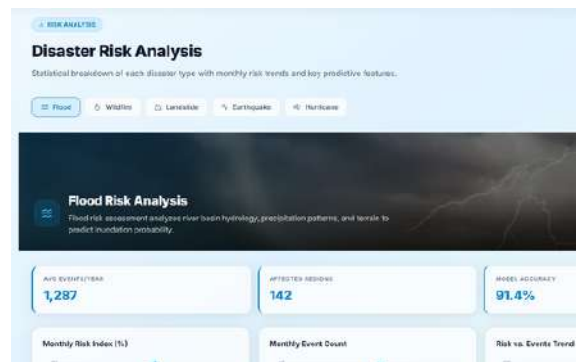


News



Prediction

Generate AI-powered disaster risk predictions for any selected location.



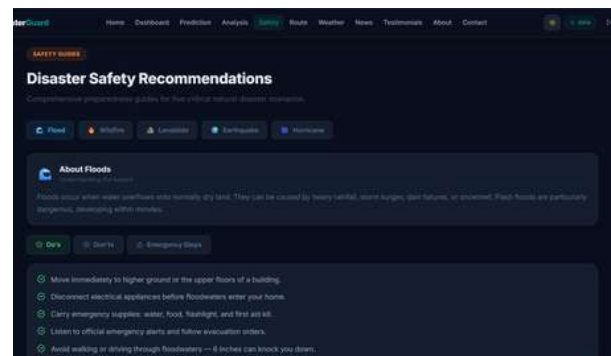
Analysis

View detailed prediction insights and supporting environmental information.



Weather

Monitor live weather conditions for better situational awareness.



Safety

Displays disaster-specific safety guidelines and precautionary measures to improve preparedness and reduce potential risks.

DEVELOPMENT CHALLENGES



DATASET PREPARATION

Collected historical disaster datasets from multiple sources and cleaned inconsistent records to create reliable data for machine learning.



API INTEGRATION

Integrated weather, mapping, geocoding, and routing APIs while managing different request formats and response structures.



MACHINE LEARNING

Evaluated and optimized Random Forest and XGBoost models to improve disaster risk prediction accuracy and overall model performance.



DEPLOYMENT

Configured the application, environment variables, and cloud services to prepare the system for reliable production deployment.



ROUTE VISUALIZATION

Implemented interactive maps and safe route generation to provide users with accurate navigation and route guidance.



FRONTEND-BACKEND COMMUNICATION

Established seamless communication between the React frontend and Flask backend using REST APIs for efficient data exchange.

EXPLORE THE PROJECT

Explore the key modules of DisasterGuard.



LIVE DEMO

Experience the deployed DisasterGuard application directly in your browser.



GITHUB REPOSITORY

Explore the complete source code, project structure, and setup instructions.



PROJECT CASE STUDY

View the complete documentation covering design, architecture, and implementation.



YOUTUBE WALKTHROUGH

Watch an in-depth explanation of the project, technologies, and development process.



LINKEDIN SHOWCASE

Discover a short project demonstration and overview shared with the professional community.

PROJECT STATUS & FINAL NOTE

Final Remarks

DisasterGuard demonstrates the integration of artificial intelligence, machine learning, real-time APIs, and modern web technologies to create a user-friendly platform for disaster awareness and risk assessment. The project reflects the practical application of AI in solving real-world challenges through an intuitive and scalable web solution.

Project Status

DisasterGuard is a functional prototype developed as a Final Year B.Tech project for educational, research, and portfolio purposes. It showcases AI-powered disaster risk prediction using historical datasets, real-time weather information, and interactive mapping technologies.

While the application demonstrates the complete development workflow and core functionalities, it is intended as a decision-support prototype and should not be considered an official disaster forecasting or emergency management system.

ABOUT

MANSI POTWAR

Computer Science & Engineering Graduate

**Aspiring Data Analyst | AI & Full-Stack
Enthusiast**

ABOUT ME

I am a Computer Science & Engineering graduate with a strong interest in Data Analytics, Artificial Intelligence, and modern web technologies. I enjoy transforming data into meaningful insights and building intelligent applications that solve real-world problems.

Through projects like DisasterGuard, I continue to strengthen my analytical thinking, problem-solving, and software development skills.

TECHNICAL INTEREST



Data Analytics



Artificial Intelligence & Machine Learning



Full-Stack Web Development



Data Visualization

CONNECT WITH ME



LinkedIn

<https://www.linkedin.com/in/mansi-potwar/>



GitHub

<https://github.com/maanasieXAI9>



Email

maanasieXAI9@gmail.com



Portfolio (Coming Soon)

**Turning ideas into intelligent
digital**



SKILLS

Languages: Python, JavaScript, TypeScript

Frontend: React.js, Tailwind CSS, Vite

Backend: Flask, REST APIs

Database: MongoDB Atlas

Data & ML: Pandas, NumPy, Scikit-learn, XGBoost,
Random Forest

Tools: Git, GitHub, VS Code, Postman

THANK YOU

THANK YOU

FOR EXPLORING DISASTERGUARD

This project has been an incredible learning experience, allowing me to combine my passion for data, technology, and problem-solving to build a solution that can make a real difference.

I hope this case study gives you valuable insights into the thought process, dedication, and technologies behind DisasterGuard.

Your feedback and suggestions are always welcome as they inspire me to continue learning and improving.

LET'S CONNECT



LinkedIn

<https://www.linkedin.com/in/mansi-potwar/>



GitHub

<https://github.com/maanasieXAI9>



Email

maanasieXAI9@gmail.com



Location

India

**Turning ideas into intelligent
digital solutions.**



BUILDING SOLUTIONS
FOR A SAFER,
MORE PREPARED
TOMORROW.

